

A Drift-Aware MLOps Framework for Multi-Model Serving: A Network Intrusion Detection Case Study

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Abstract. Machine learning systems deployed in production can degrade when data distributions change, even if source code remains unchanged. This issue is critical for network intrusion detection, where traffic and attack behavior evolve over time. This paper does not propose a new intrusion detection classifier; instead, it presents a baseline drift-aware MLOps framework for user-imported and training-job-created tabular models, using Network Intrusion Detection Systems (NIDS) as a case study. The framework separates model governance from model serving through a control-plane/data-plane architecture and formalizes the lifecycle over tenants, models, inference requests, production windows, drift scores, and lifecycle actions. The prototype uses Django/FastAPI services, PostgreSQL, Redpanda, Kubernetes-native training through Argo Workflows and Kubeflow PyTorchJob resources, S3-compatible object storage, optional MLflow support, and Evidently AI. It demonstrates how model artifacts can be onboarded, registered in a native model-evolution registry, served as APIs, monitored from production logs, and evaluated for Data Drift.

Keywords: MLOps · Data Drift · Multi-model Serving · Model Monitoring · Network Intrusion Detection

1 Introduction

Machine learning (ML) models are increasingly deployed as continuously running services rather than offline experimental artifacts. In a production setting, a model receives new input samples over time, generates predictions for downstream users or systems, and depends on monitoring mechanisms to remain trustworthy. Unlike traditional software services, an ML service may degrade even when its source code and infrastructure remain unchanged. One major cause is Data Drift, where the production data distribution differs from the reference distribution used during training, validation, or initial deployment.

This problem is especially relevant in Network Intrusion Detection Systems (NIDS). Network traffic is non-stationary: benign usage patterns change, infrastructure evolves, and attack strategies continuously adapt. A classifier trained

on historical flow features may therefore become less reliable when deployed in a new environment or under a new attack distribution. An operational NIDS should not only expose a prediction API but also record production data, compare it with reference data, and trigger lifecycle actions when the monitored distribution changes.

Existing MLOps research provides architectural concepts for automating ML development, deployment, and monitoring [1]. Other works discuss MLOps maturity, pipeline platforms, data quality, and continual model maintenance [4, 7, 8]. However, practical systems often treat monitoring as a passive observability layer. In many prototypes, drift reports are generated after deployment, but the monitoring result is not explicitly modeled as a control signal for model lifecycle decisions.

This paper presents a baseline framework that connects multi-model serving with drift-aware monitoring. The goal is to describe a reusable MLOps framework, not to claim a new state-of-the-art NIDS classifier. In the proposed design, NIDS is used as a case study because it naturally exposes the need for drift-aware operation. The framework is intentionally scoped to tabular classification models that are either imported by users or produced by controlled training jobs and then converted into serving-compatible artifacts.

The main contributions of this paper are:

- We propose a tenant-aware and drift-aware MLOps framework that integrates model onboarding, dynamic serving, production logging, drift detection, and lifecycle adaptation triggers.
- We formalize the framework using tenants, model functions, reference data, production windows, feature-level drift indicators, dataset-level drift ratio, and an action-selection function.
- We implement a prototype platform and instantiate it with a Network Intrusion Detection case study to demonstrate the feasibility of the proposed lifecycle.

The remainder of this article is structured as follows. Section 2 reviews related work on MLOps lifecycles, serving platforms, data quality, drift monitoring, and ML-based NIDS. Sections 3–5 define the problem, present the framework, and describe the drift-aware lifecycle algorithm; Sections 6–8 instantiate the framework in the NIDS case study, describe the prototype, and outline the evaluation design. Sections 9 and 10 discuss limitations and conclude the paper.

2 Related Work

2.1 MLOps Architectures and Lifecycle

MLOps extends DevOps principles to ML systems by coordinating data engineering, experimentation, training, deployment, monitoring, and governance. Kreuzberger *et al.* define MLOps as a paradigm for reliably and efficiently developing and maintaining ML systems in production [1], while John *et al.* propose

a maturity model for assessing MLOps adoption [4]. Sculley *et al.* show that ML systems accumulate technical debt beyond model code, including data dependencies, configuration, monitoring, and feedback loops [2]. Amershi *et al.* further show that software engineering for ML differs from traditional software engineering because data, models, and code evolve together [3]. Our work narrows these lifecycle concerns to drift-aware signals for tenant-aware multi-model serving.

2.2 MLOps Platforms, Model Registry, and Model Serving

Implementation-oriented MLOps studies show how platform components can connect model development and deployment. Zhou *et al.* discuss a machine learning pipeline platform [5], and Subramanya *et al.* apply an MLOps workflow to electricity market forecasting [7]. TFX demonstrates how production ML platforms can integrate pipeline stages such as data ingestion, validation, training, and serving [6]. MLflow provides experiment tracking and model lifecycle support [12]. Separately, Clipper treats online prediction serving as a systems problem involving model abstraction and serving infrastructure [13]. In contrast to systems that focus on a single part of the lifecycle, this baseline combines registry metadata, generated endpoints, production logs, and drift signals in one prototype workflow.

2.3 Data Quality, Data Drift, and Concept Drift Monitoring

Production ML reliability depends on the data observed after deployment as well as on the pipelines that manage that data. Renggli *et al.* argue that data quality should be treated as a first-class MLOps concern [8], and Polyzotis *et al.* identify data management challenges that arise in production ML pipelines [9]. Data Drift and concept drift are related but distinct concerns: Gama *et al.* survey adaptation to changing data streams [10], while Webb *et al.* characterize different forms of concept drift [11]. In this paper, Data Drift is used as an operational signal. Because production labels may be delayed or unavailable, the system first compares input distributions and then triggers inspection or workflow handoff when drift is detected.

2.4 Cybersecurity/NIDS as a Non-Stationary Case Study

Network intrusion detection provides a useful non-stationary case study for drift-aware MLOps. CIC-IDS2017 provides labeled intrusion traffic for evaluating intrusion detection models [15], but real deployment environments can differ from dataset conditions. Sommer and Paxson argue that ML-based intrusion detection faces challenges outside closed-world assumptions, including deployment context, generalization, and evolving traffic distributions [16]. This paper therefore uses NIDS to illustrate lifecycle and monitoring requirements, not to claim a new state-of-the-art intrusion detection classifier.

2.5 Research Gap

The reviewed literature covers MLOps architectures, production ML platforms, model serving, data quality, drift adaptation, and NIDS deployment challenges. However, these strands are often treated separately. This paper addresses the gap by presenting a baseline, tenant-aware, multi-model serving framework in which drift monitoring is represented as an explicit lifecycle signal connected to registry metadata, serving endpoints, production logs, and review-driven adaptation actions.

3 Problem Formulation

Let $\mathcal{U} = \{u_1, u_2, \dots, u_N\}$ denote the set of tenants using the platform. Each tenant $u \in \mathcal{U}$ owns a set of registered models:

$$\mathcal{M}_u = \{m_{u,1}, m_{u,2}, \dots, m_{u,K_u}\}. \quad (1)$$

Each model $m_{u,i}$ is represented by an inference function parameterized by θ :

$$\hat{y}_t = f_{\theta}^{u,i}(x_t), \quad (2)$$

where $x_t \in \mathbb{R}^d$ is the input vector at time t , d is the number of input features, and $\hat{y}_t$ is the prediction. For a tabular classification model, $\hat{y}_t$ may be a class label or a probability vector over class labels.

For every inference request, the platform records a production log:

$$z_t = (u, i, x_t, \hat{y}_t, c_t, t), \quad (3)$$

where c_t is the prediction confidence if available. The log stream associated with a tenant-model pair (u, i) forms the basis for monitoring.

Each deployed model has a reference dataset:

$$D_{\text{ref}}^{u,i} = \{x_1, x_2, \dots, x_n\}, \quad (4)$$

which represents the distribution observed during training, validation, or initial acceptance. At monitoring time T , the current production data window is:

$$D_{\text{cur}}^{u,i}(T) = \{x_t \mid T - W \leq t \leq T\}, \quad (5)$$

where W is the monitoring window size. The objective is not only to compute predictions but also to select a lifecycle action:

$$a_T = \pi(m_{u,i}, D_{\text{ref}}^{u,i}, D_{\text{cur}}^{u,i}(T), S_T), \quad (6)$$

where $\pi(\cdot)$ is the adaptation policy and S_T is the current lifecycle state of the model. Possible actions include no operation, alert generation, manual review, retraining trigger, and review-driven rollback or promotion candidates.

For each feature j , the framework computes a drift score:

$$\delta_j = \Delta(P_{\text{ref}}(x_j), P_{\text{cur}}(x_j)), \quad (7)$$

where $\Delta(\cdot)$ is a statistical distance or hypothesis test between the reference and current feature distributions. A feature-level drift indicator is defined as:

$$I_j = \begin{cases} 1, & \delta_j > \tau_j, \\ 0, & \text{otherwise,} \end{cases} \quad (8)$$

where τ_j is the threshold for feature j . The dataset-level drift ratio is:

$$\rho = \frac{1}{d} \sum_{j=1}^d I_j. \quad (9)$$

The final drift decision is:

$$\text{Drift}(D_{\text{ref}}, D_{\text{cur}}) = \begin{cases} 1, & \rho \geq \tau_D, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

where τ_D is the dataset-level threshold. This formulation separates model prediction from monitoring and makes Data Drift an explicit input to lifecycle decisions.

4 Proposed Framework

The logical flow of the proposed framework consists of four stages. First, a user onboards a model artifact or registers an artifact produced by a controlled training workflow through the control plane. Second, the data plane exposes the model as a prediction endpoint. Third, each prediction produces an asynchronous log event. Fourth, the monitoring layer periodically reconstructs the current production window and compares it with the reference data. If drift is detected, the adaptation layer creates an alert or triggers a retraining workflow.

4.1 Control Plane

The control plane manages user-facing and governance operations. It stores tenants, authentication credentials, API keys, model metadata, artifact locations, endpoint URLs, access modes, and model statuses. In the prototype, this layer is implemented using Django. A model record contains the model name, description, owner tenant, artifact URI, access mode, generated endpoint, and lifecycle status.

When a model artifact is uploaded or registered from a completed training job, the control plane validates the available metadata, stores artifact references, and records the model in the native registry. The model can then be assigned an endpoint of the form `/models/{model_id}/predict`. This design decouples model ownership and metadata management from online inference.

4.2 Data Plane

The data plane is implemented as a generic FastAPI model server. The logical serving path resolves models through a shared API, while deployment adapters may create dedicated runtime resources when configured. The server reads the corresponding model metadata, loads the model artifact through MLflow pyfunc, validates the request payload against the expected input format when possible, and returns the prediction result.

Tenant-aware access control is enforced before inference. Public models can be invoked directly, while private models require a valid API key or bearer token. The resolved tenant identifier must match the model owner. This mechanism supports a platform setting where multiple users upload and serve independent models through a shared serving layer.

4.3 Production Logging Layer

Online inference should not depend synchronously on database writes. Therefore, the framework decouples prediction from logging through a message broker. After producing a prediction, the model server emits a production log event to Redpanda. A consumer service batches log events and persists them to PostgreSQL. Each log stores the tenant identifier, model identifier, timestamp, feature payload, and prediction result.

This design has two advantages. First, it keeps monitoring writes outside the synchronous prediction path. Second, it creates a unified production data source that can be queried per tenant and per model for drift detection, audit, and evaluation.

4.4 Drift Monitoring Layer

The monitoring layer compares reference data with recent production data. In the prototype, Evidently AI is used to compute feature-level drift and dataset-level drift summaries. Reference data can be stored as object storage artifacts, while production data are reconstructed from PostgreSQL logs. Drift reports can be exported as HTML or JSON artifacts for inspection.

4.5 Adaptation Layer

The adaptation layer maps monitoring outputs to lifecycle actions. In the baseline implementation, the primary action is a drift alert or workflow trigger. The framework can be extended to support retraining jobs, model evaluation gates, review-driven promotion, rollback, and notification workflows. This distinction is important: the current prototype demonstrates a drift-aware control loop, while fully autonomous retraining and deployment remain future work.

Require: Model package or training artifact A , metadata H , tenant u
Ensure: Registered model endpoint e

- 1: Validate package extension and archive structure
- 2: Validate serving metadata or record deployability status
- 3: Store or reference A in artifact storage
- 4: Create model record $m_{u,i}$ in the control plane
- 5: Assign lifecycle state $S \leftarrow \text{validated}$
- 6: Generate endpoint $e \leftarrow \text{/models/\{i\}/predict}$
- 7: Return endpoint e

Fig. 1. Model onboarding procedure.

5 Drift-Aware Lifecycle Algorithm

The model lifecycle can be represented as a state transition system:

$$S \in \{\text{uploaded, validated, deployed, monitored, drifted, reviewing, retraining, promoted, failed}\}. \quad (11)$$

The transition function is:

$$S_{T+1} = \Phi(S_T, m_{u,i}, \rho, \tau_D, a_T), \quad (12)$$

where $\Phi(\cdot)$ updates the model state according to the drift ratio and the selected action. If $\rho < \tau_D$, the model remains in the monitored state. If $\rho \geq \tau_D$, the model is marked as drifted and the system can trigger alerting, review, or a workflow handoff.

6 Case Study: Network Intrusion Detection

The NIDS model is used as a case study to instantiate the proposed framework. A network flow is represented as a tabular feature vector $x \in \mathbb{R}^d$ containing statistics such as duration, packet counts, byte rates, and protocol-related attributes. The model predicts whether the observed traffic is benign or belongs to an attack class.

In the current prototype, XGBoost is used as the classification algorithm because it is widely used for tabular data and can be packaged into a serving-compatible model artifact. The project contains multiple trained model versions, including binary-classification artifacts used for upload and model-evolution validation. These models are treated as artifacts and registry versions rather than hard-coded services. This allows the case study to demonstrate the general framework behavior: model registration, endpoint invocation, production logging, and drift monitoring.

NIDS is an appropriate case study for drift-aware MLOps for three reasons. First, network environments change over time, causing benign traffic distributions to shift. Second, attack patterns evolve, making historical training data

Require: Input x_t , model identifier i , credential q
Ensure: Prediction $\hat{y}_t$ and optional lifecycle action a_T

- 1: Authenticate q and resolve tenant u
- 2: Retrieve model metadata $m_{u,i}$
- 3: Load or reuse cached model $f_{\theta}^{u,i}$
- 4: Compute prediction $\hat{y}_t \leftarrow f_{\theta}^{u,i}(x_t)$
- 5: Emit log event $z_t = (u, i, x_t, \hat{y}_t, t)$
- 6: **if** monitoring condition is satisfied **then**
- 7: Build production window $D_{\text{cur}}^{u,i}(T)$
- 8: **for** each feature $j = 1$ to d **do**
- 9: Compute drift score δ_j
- 10: Compute indicator I_j
- 11: **end for**
- 12: Compute drift ratio ρ
- 13: Select action $a_T \leftarrow \pi(m_{u,i}, D_{\text{ref}}, D_{\text{cur}}, S_T)$
- 14: **if** $\rho \geq \tau_D$ **then**
- 15: Trigger alert, review, or workflow handoff
- 16: **end if**
- 17: **end if**
- 18: Return $\hat{y}_t$

Fig. 2. Drift-aware inference and monitoring procedure.

incomplete. Third, labeled production data are often delayed or unavailable, so input distribution monitoring can provide an early warning signal before supervised performance metrics are available.

7 Prototype Implementation

This section describes the implemented prototype as reflected in the repository. The purpose is to document the system boundary and implementation coverage, not to claim production readiness or performance characteristics.

7.1 Prototype Architecture

The prototype follows a control-plane/data-plane architecture. The control plane is implemented with Django and Django REST Framework and stores users, tenants, training jobs, model API records, registry versions, deployment metadata, and drift monitoring records in PostgreSQL. The data plane is implemented as a FastAPI model server that resolves a model endpoint, loads the serving artifact, performs prediction, and emits production log events. Redpanda is used as the asynchronous log transport, and a consumer service persists prediction records into PostgreSQL for later drift analysis.

The implementation also includes S3-compatible artifact storage, Argo-triggered Kubernetes training workflows using Kubeflow PyTorchJob resources, a Python training runner, a model packager, MLflow integration, Evidently-based drift

monitoring, and a Native Registry/Model Evolution layer. In the user-facing workflow, Native Registry data stored in PostgreSQL is the source of truth for Model Evolution. MLflow is treated as internal experiment lineage and model-format support rather than a required user interface.

Table 1. Prototype modules and responsibilities.

Module	Implementation	Responsibility
Control plane	Django/DRF	Manages tenants, jobs, registry, deployment, and drift APIs.
Training execution	Argo + Kubeflow PyTorchJob + runner	Executes configured entry points and returns artifacts.
Artifact/registry layer	Object storage + PostgreSQL	Stores model artifacts, metadata, and version summaries.
Packaging/serving	Packager + FastAPI	Prepares or loads serving artifacts and exposes prediction endpoints.
Logging/drift	Redpanda + consumer + Evidently	Persists production logs and stores drift summaries.
Deployment configuration	Docker Com- pose/K8s/Argo files	Supports local integration and prototype/testbed deployment.

7.2 Training Job Execution

Training jobs can be created from uploaded source packages or from code stored through the Source Editor workflow. In the Kubernetes-native path, the control plane validates the selected model source and reference data, prepares S3-compatible source and data references, creates presigned input and output URLs, and sends a training request to Argo Events. Argo Workflows then instantiates a Kubeflow `PyTorchJob` resource in the configured training namespace. The training container executes the configured entry point, writes model files and optional metadata to the expected output directories, and uploads the resulting `model.tar.gz` artifact to object storage.

The non-local training path does not require the user script to import MLflow or receive MLflow credentials. Instead, the user script writes normal model artifacts and optional metadata files, and the runner is responsible for packaging the output in a format the control plane can ingest. After the workflow completes, a webhook updates the `TrainingJob` record with artifact locations; the control plane can then ingest tracking metadata and synchronize Native Registry and `ModelVersion` records when the trained artifact is registered as a model.

7.3 Training Runner Output Contract

After training completes, the runner produces a required `model.tar.gz` archive and, when available, an `_mlops` metadata bundle containing training summary, metrics, parameters, model insights, artifact manifest, stdout/stderr logs, and warnings. Only the archive is required; the remaining metadata improves registry display and version comparison but does not invalidate the artifact when absent.

7.4 Metadata Ingestion and Native Registry

When a completed job is ingested, the control plane downloads the artifact, extracts metadata, and stores metrics, parameters, insights, manifest, and deployability summaries. These summaries are mirrored into Model Evolution when a model is registered from the job.

The Native Registry groups versions through `ModelFamily` and `ModelVersion` records linked to deployable `ModelAPI` records. Model Evolution displays tracking status, metrics, parameters, artifacts, model insights, deployability state, version comparison, deployment actions, drift summaries, and basic routing aliases.

7.5 Model Packaging and Serving

The prototype includes a model packager that prepares serving-compatible packages or images for supported artifacts. Build execution can be triggered through a local Docker path or through an Argo-oriented workflow path, depending on deployment configuration. The FastAPI model server loads the model artifact, including MLflow pyfunc artifacts where available, and exposes generated prediction endpoints. Prediction requests can be public or private depending on model configuration and are logged asynchronously for monitoring.

No latency or throughput claims are made in this paper. The current implementation demonstrates the integration path from registry metadata to serving endpoint and production log generation.

7.6 Drift Monitoring

Drift monitoring uses production records stored in PostgreSQL and reference data associated with a model. The Evidently integration computes feature-level and dataset-level drift summaries and stores report locations and drift result metadata in the control plane. The drift result can be exposed as a report link or summarized in Model Evolution.

In this baseline, drift is a monitoring and adaptation signal. It does not automatically redeploy a model, perform rollback, or select traffic weights. Such decisions require evaluation gates and human review policies that are outside the current baseline.

7.7 MLflow Tracking Integration

MLflow is used as optional internal tracking. After the control plane ingests the training artifact, it may log metrics, parameters, and metadata artifacts to an MLflow tracking server and store the run identifier. If MLflow tracking is disabled, Model Evolution still displays metrics, parameters, artifacts, model insights, and deployability information from the Native Registry. If MLflow logging fails, the metadata retained in PostgreSQL remains available for review and deployment decisions.

This design avoids requiring user training scripts to import MLflow. It also avoids making the MLflow UI the primary product workflow. The user-facing Model Evolution pages are backed by Native Registry records.

7.8 Deployment and Operational Scope

The repository contains Docker Compose configuration for local integration and Kubernetes-oriented manifests for a K3s-based prototype environment. It also includes CI/CD configuration for validation, container build, image publication, and GitOps-style deployment configuration updates. These assets document deployment-oriented automation but are reported as prototype support rather than production hardening or evidence of the latest workflow success.

The AWS-facing boundary supports object storage and Kubernetes-native training execution where configured. Conceptually, the web frontend calls the Django/DRF control plane for model upload, training, registry, deployment, and drift views; the control plane exchanges artifacts with object storage and submits training requests to Argo Events; Argo Workflows creates KubeFlow PyTorchJob resources for training pods; the runner returns artifacts for Native Registry ingestion; the FastAPI server handles inference and emits logs to Redpanda; the consumer stores logs in PostgreSQL; and the drift service writes summaries back to the control plane. When the cluster is configured with Karpenter, pending training pods can match training-specific node pools and AWS EC2 node classes, and empty training nodes can be consolidated according to node-pool policy. Runtime evidence such as KubeFlow/Argo run logs, pod status, artifact uploads, registry ingest records, Karpenter NodePool or NodeClaim events, health checks, and screenshots should be attached separately before stronger operational claims.

The implemented baseline covers authentication, manual model upload, Kubernetes-native training submission, metadata ingestion, Model Evolution views, version comparison, build/deploy/smoke-test actions, production logging, drift report generation, and a basic routing alias. Its limitations are intentionally explicit: autonomous retraining, automated promotion, canary or blue-green deployment, weighted traffic routing, resource quota enforcement, billing, and production-grade hardening are outside the completed baseline. K3s/Kubernetes deployment, KubeFlow training runs, Karpenter provisioning behavior, and CI/CD status should be reported with separate runtime evidence.

8 Evaluation Design

Because this paper is a baseline, evaluation focuses on feasibility and system behavior rather than NIDS accuracy. The validation plan covers five aspects: model onboarding, inference serving, production logging, drift detection, and operational workflow reachability. Onboarding checks whether model packages or completed training artifacts can be validated, stored, registered, and assigned endpoint metadata. Serving and logging checks whether tabular requests produce predictions and queryable production logs. Drift validation compares reference and production windows and reports drifted features, drift ratio, dataset-level decision, and report artifacts. Operational validation should use Kube-flow/Argo run status, training pod logs, artifact storage paths, registry ingest results, container image references, Kubernetes status, health checks, Karpen-ter node-provisioning evidence where configured, and end-to-end training or inference traces when runtime evidence is available.

9 Discussion

The proposed framework reframes MLOps monitoring as part of an adaptive control loop. The model server does not only return predictions; it also emits production evidence. The monitoring layer does not only generate reports; it produces a signal that can affect model lifecycle decisions. This design is useful for a SaaS or PaaS-style environment where different users upload different models but still need a common operational pipeline.

The current scope is intentionally limited. First, the system supports tabular classification models that can be represented as serving-compatible artifacts. Image, text, time-series, and large language models require additional schema validation and monitoring metrics. Second, arbitrary user-uploaded model artifacts introduce security risks. A hardened system must isolate execution, validate dependencies, restrict archive extraction, and enforce resource quotas. Third, autonomous retraining and promotion require reliable evaluation gates, rollback policies, and human approval mechanisms. Therefore, this baseline treats drift detection as an adaptation signal rather than an unconditional redeployment decision.

10 Conclusion

This paper presented a baseline design for a drift-aware MLOps framework for multi-model serving. The framework integrates tenant-aware model onboarding, training-job execution, native model-evolution metadata, dynamic endpoints, asynchronous production logging, and reference-production drift monitoring. The system was formalized using tenants, model functions, production windows, drift indicators, lifecycle states, and adaptation actions. A Network Intrusion Detection model was used as a non-stationary security case study. Future work will focus on artifact-execution hardening, runtime deployment evidence, evaluation gates, and additional model types.

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